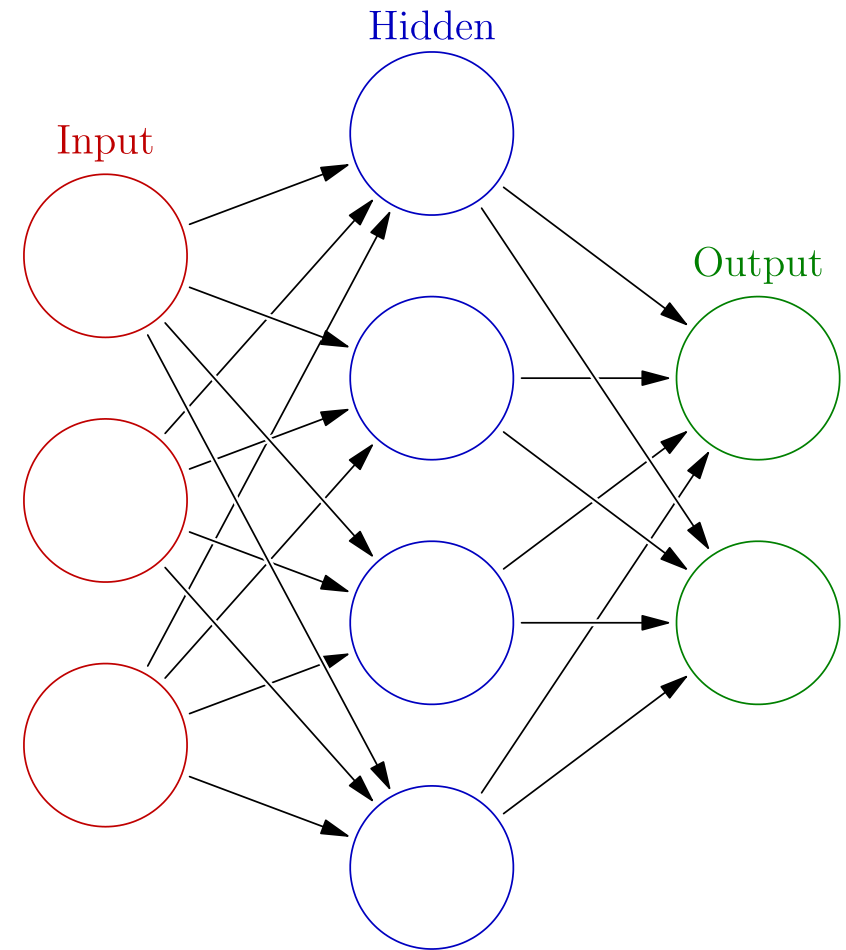


Clustering Neurons

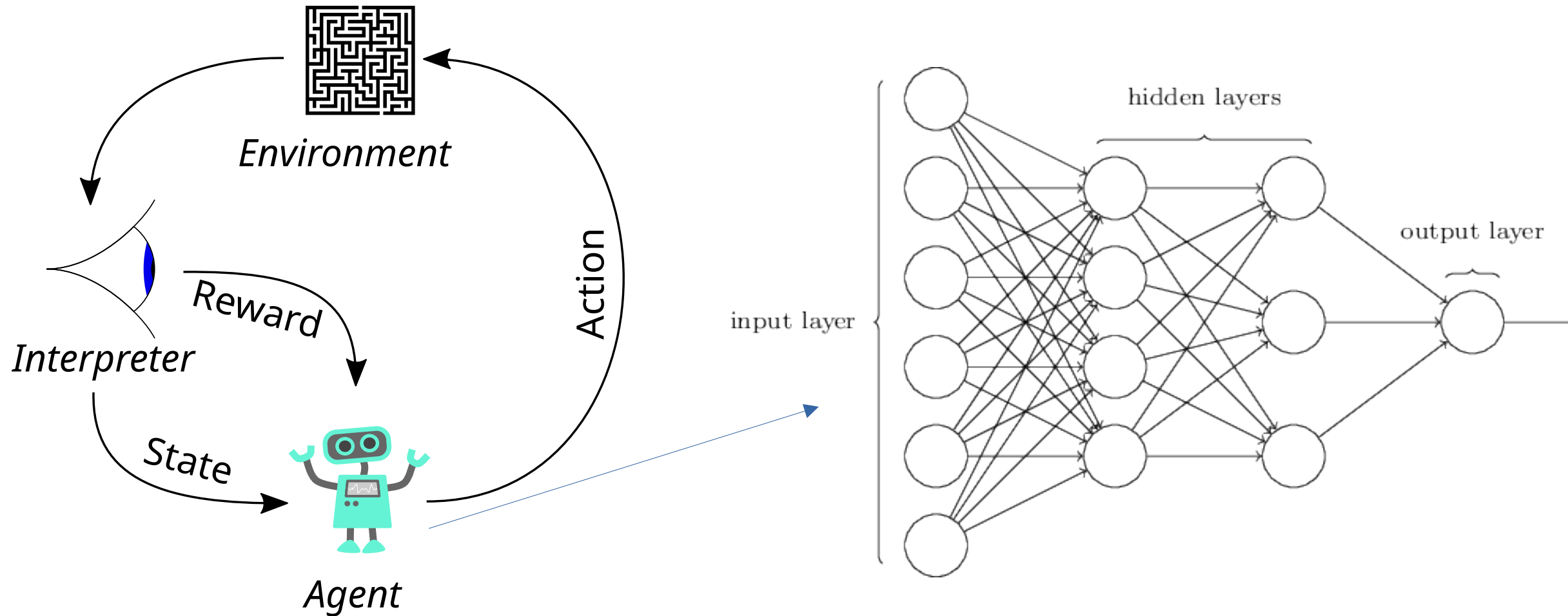
Goal: Cluster artificial neurons based on their activation pattern

How:

- Record the artificial model neuronal activations for each step of multiple SSAE tasks
- Preprocessing :
 - Standardization
 - Perform *dimensionality reduction* (multiple methods)
- ...



Recording Activations



Each iteration of the loop is a *step* in our SSAE task

Recording Activations

- Make the model solve SSAE episodes and record at each step all of it's neuron activations

$$\mathbb{R}^{N \times D}$$

- N is the total number of steps
- D is the number of neurons

Recording Activations (intuition)

$$\mathbb{R}^{N \times D}$$

Intuition: Each neuron is represented by its activations at each step during the episodes.

Reversely, each step is represented by the activation of all neurons in the neural network at that step.

Dimensionality Reduction

Why: Helps with analysis and clustering

Goal: Reduce the dimensionality of our data while retaining important information

$$\mathbb{R}^{N \times D} \mapsto \mathbb{R}^{n \times D}$$

- n is the reduced «steps» dimension ($n < N$)
- The resulting matrix is called the embeddings

Dimensionality Reduction

- **PCA**
 - Linear method,
 - Interpretable but limited to linear relationships
 - Deterministic
- **UMAP**
 - Non-linear method
 - Preserves local and global structure
 - Stochastic
- **Autoencoders**
 - Also non-linear
 - Learn representations via neural-networks
 - Higher computational cost

What to do next

$$\mathbb{R}^{n \times D}$$

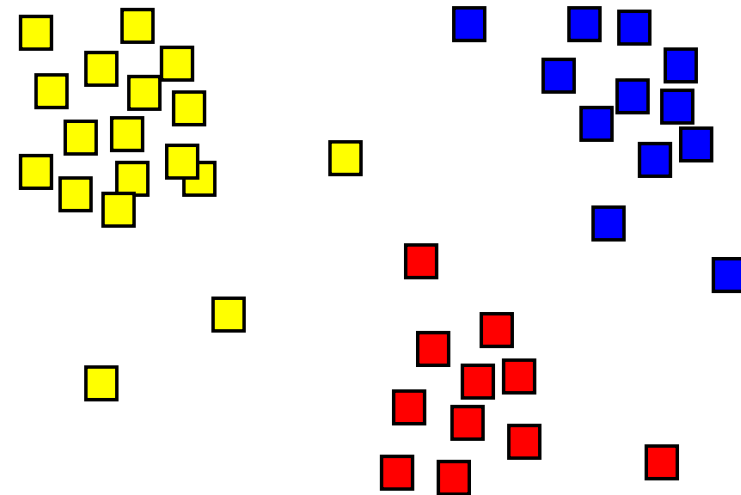
- 1) Clustering directly the embeddings
- 2) First compute the RSA matrix to then make clusters

Clustering Algorithms

Goal: Automatically detect groups/patters in our data, i.e. partitioning our data

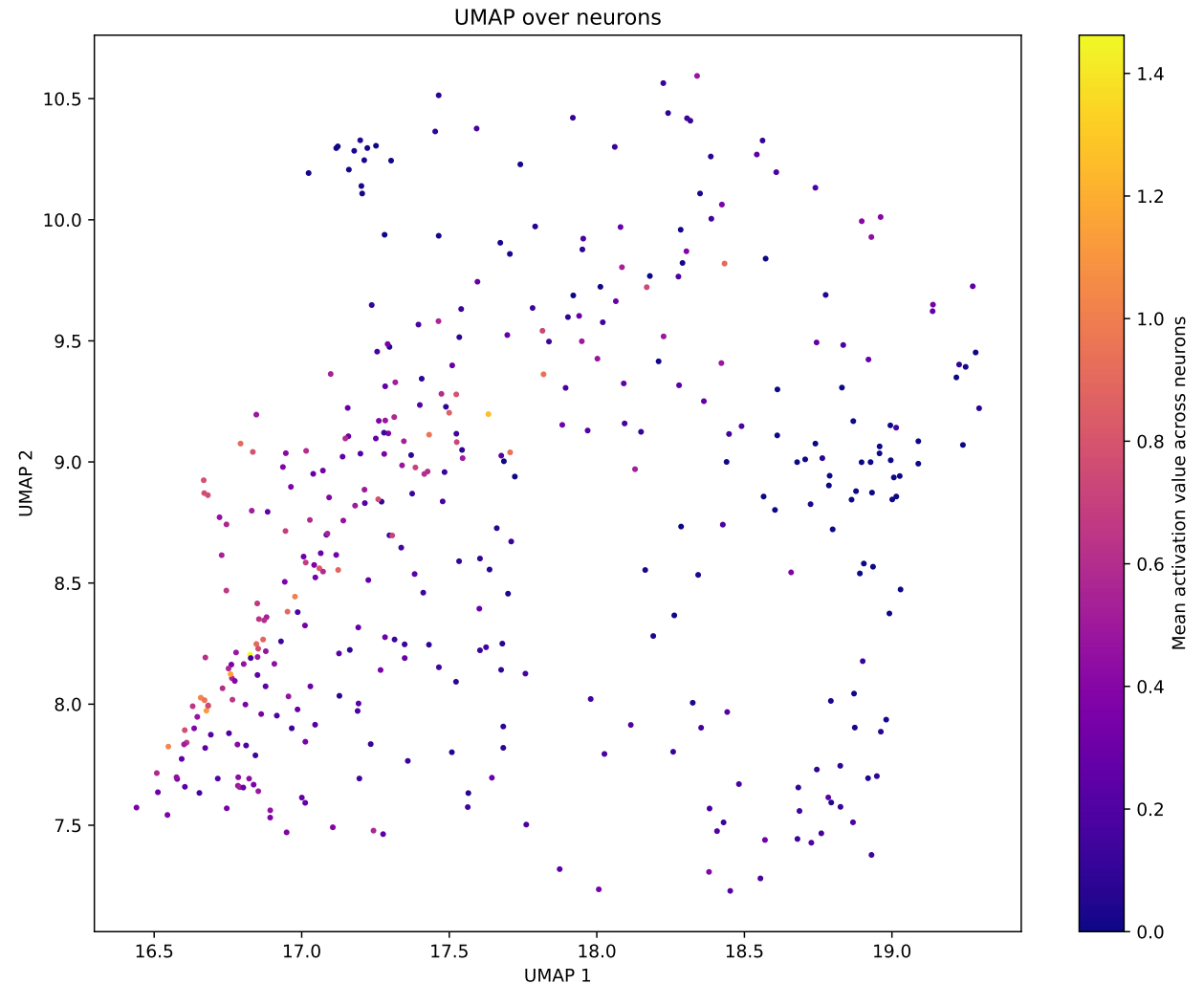
How: Multiple algorithms :
- cmeans, hierarchical clustering, (H)DBSCAN

$$\mathbb{R}^{n \times D}$$



Clustering the embeddings (1st method)

Problem: Embeddings made from UMAP are not always forming clear groups/patterns : not easily clusterable/separable



Clustering the embeddings (1st method)

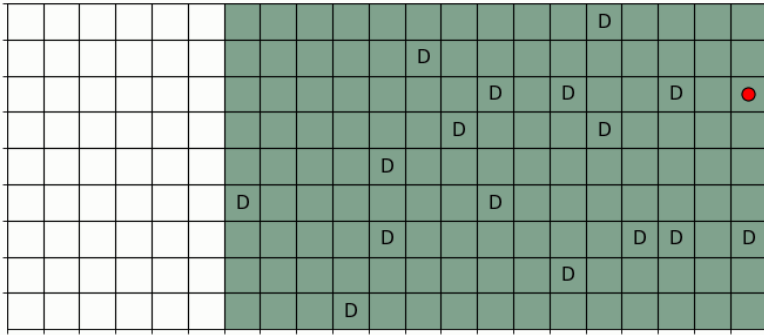
Possible solution

Refine the process by not representing the neurons by their activations in all steps but only by steps from a certain moment in the episode.

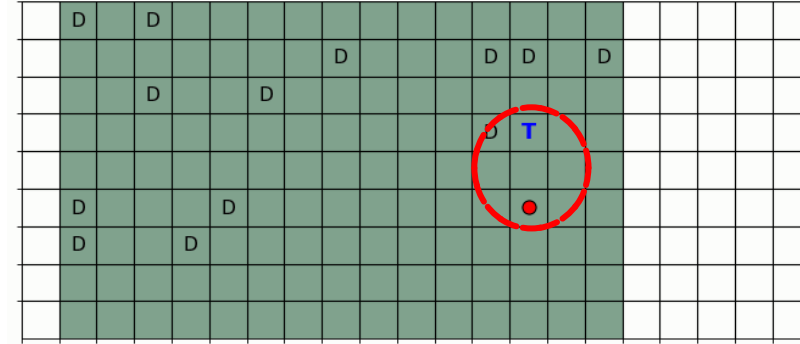
How

By clustering the steps into clusters (groups) that we will call env-clusters

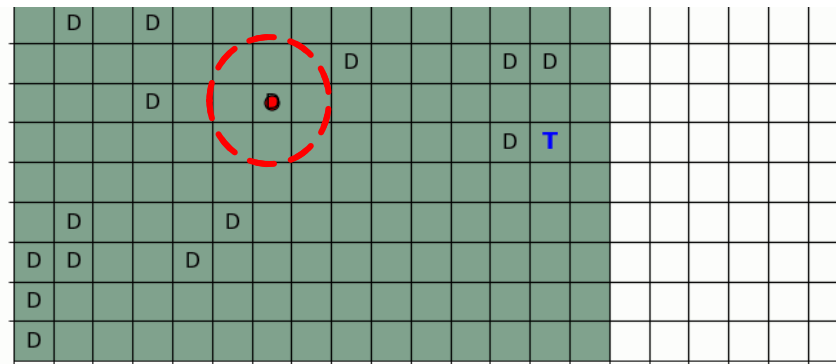
Examples of env-clusters



Hidden target



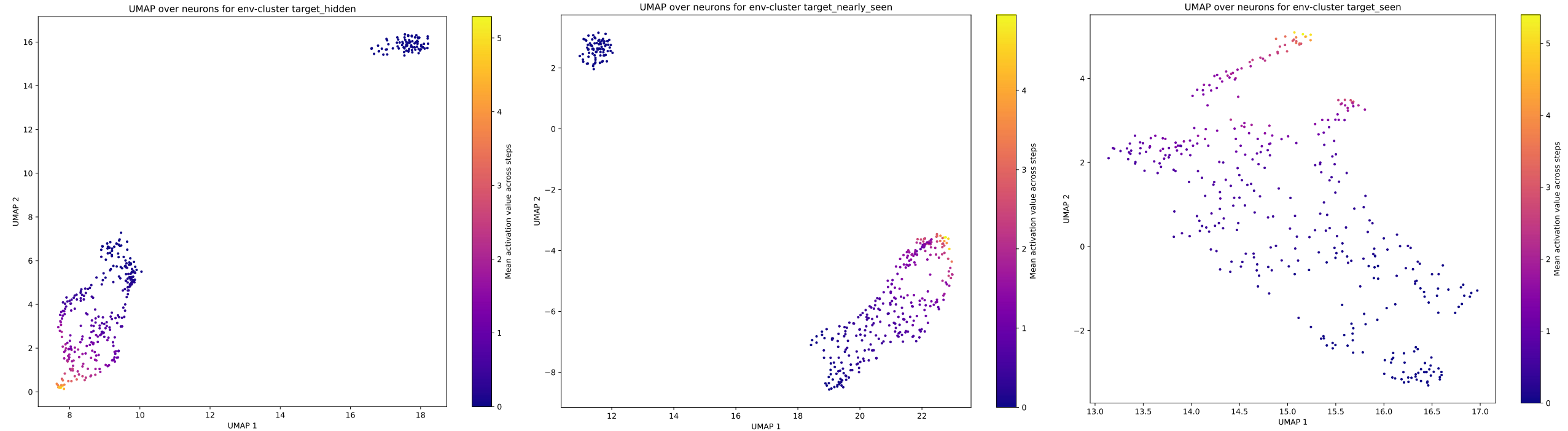
Target seen



Target nearly seen

Clustering the embeddings (1st method)

We can now cluster the neurons per env-clusters



Clustering the embeddings (1st method)

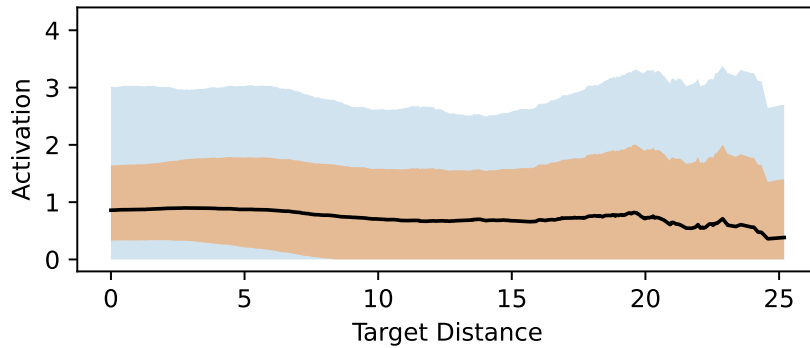
Goal: Transform 3 sets of neuron clusters into 1

We compare and combine via different processes these neuron clusters per env-cluster into one single set of neuron clusters

How: Co-association matrices, graphs, connected components

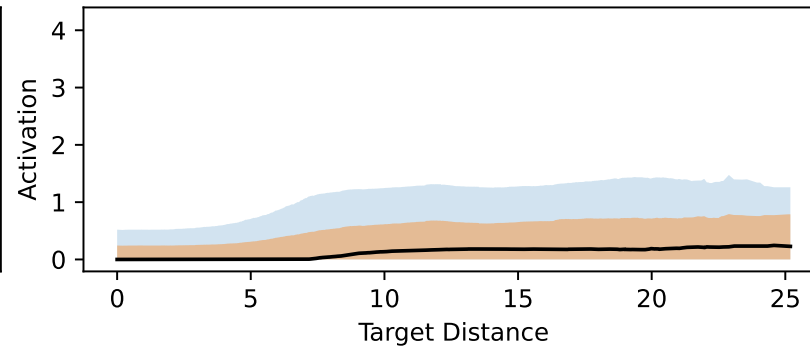
Visualizing neuron clusters

Neuron Cluster: 0

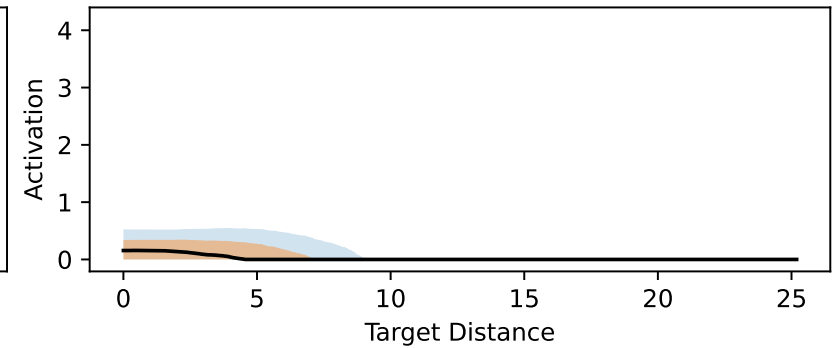


Rolling Neuron Activation vs Target Distance

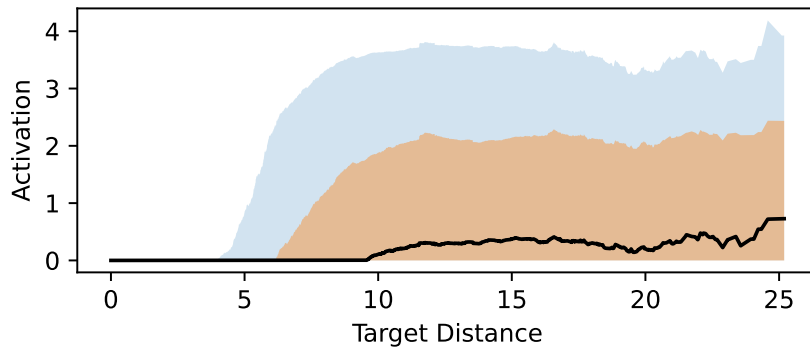
Neuron Cluster: 1



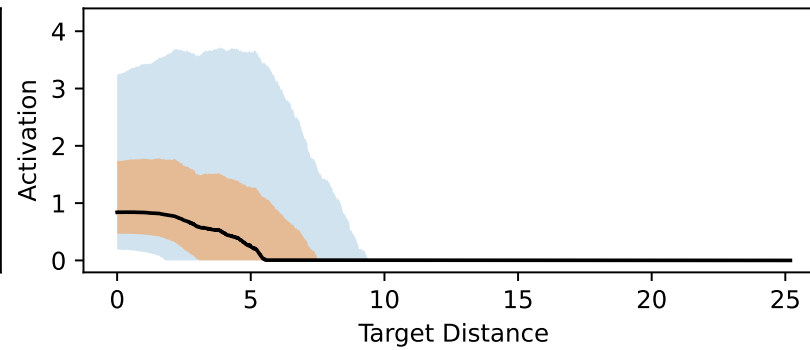
Neuron Cluster: 2



Neuron Cluster: 3



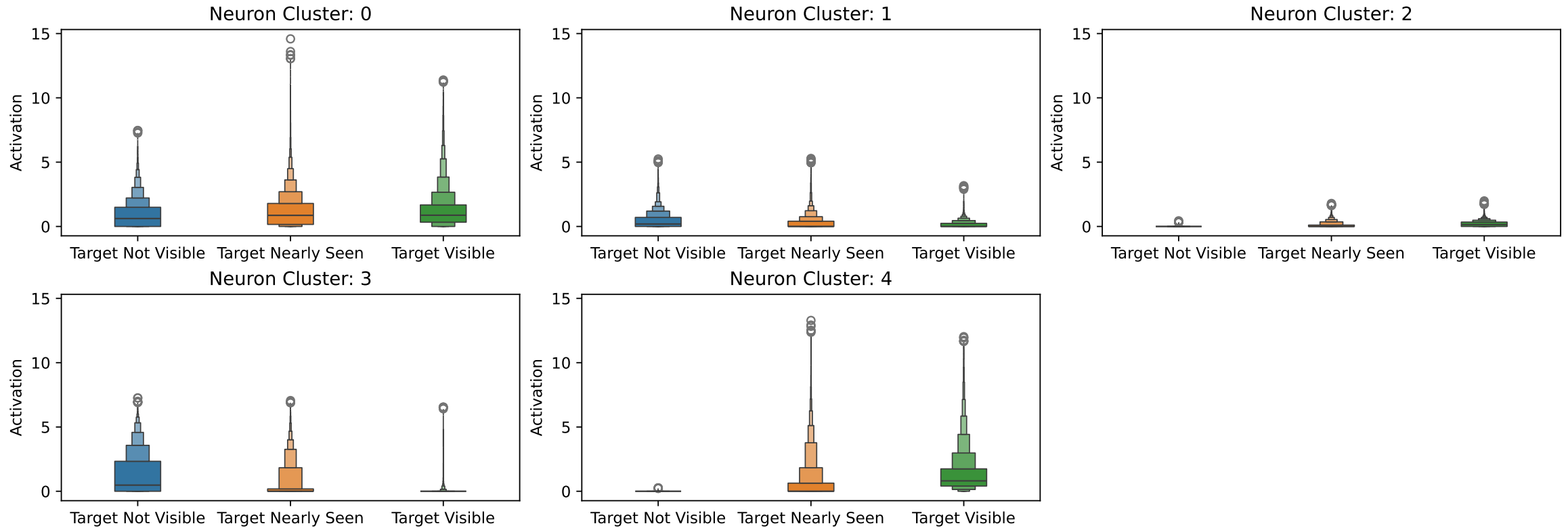
Neuron Cluster: 4



— Median 10-90% 25-75%

Visualizing neuron clusters

Neuron Activation Distribution vs Target Visibility



RSA clustering (2nd method)

Why: More grounded in literature

A Correlation-Matrix-Based Hierarchical Clustering Method for Functional Connectivity Analysis

Xiao Liu^{1,2,*}, Xiao-Hong Zhu¹, Peihua Qiu², and Wei Chen¹

¹Center for Magnetic Resonance Research, Department of Radiology, University of Minnesota, Minneapolis, Minnesota, USA

²Department of Statistics, University of Minnesota, Minneapolis, Minnesota, USA

Abstract

In this study, a correlation matrix based hierarchical clustering (CMBHC) method is introduced to extract multiple correlation patterns from resting-state functional magnetic resonance imaging (fMRI) data. It was applied to spontaneous fMRI signals acquired from anesthetized rats, and the results were then compared with those obtained using independent component analysis (ICA), one of the most popular multivariate analysis method for analyzing spontaneous fMRI signals. It was demonstrated that the CMBHC has a higher sensitivity than the ICA, particularly on a single run data. for identifying correlation structures with relatively weak connections. for instance, the

<https://doi.org/10.1038/s42003-025-07704-w>

Article

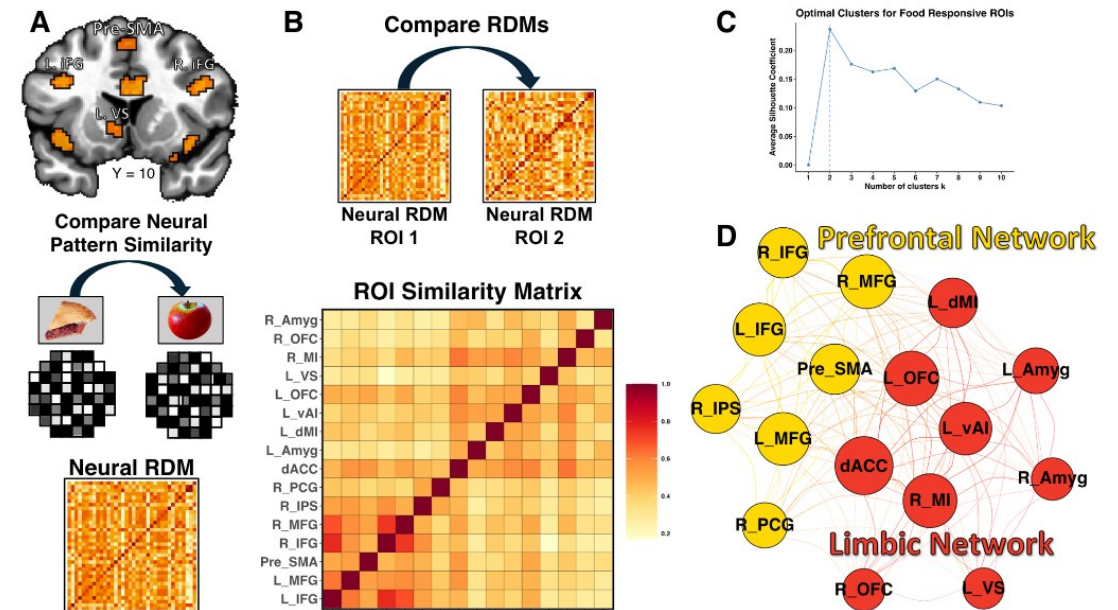


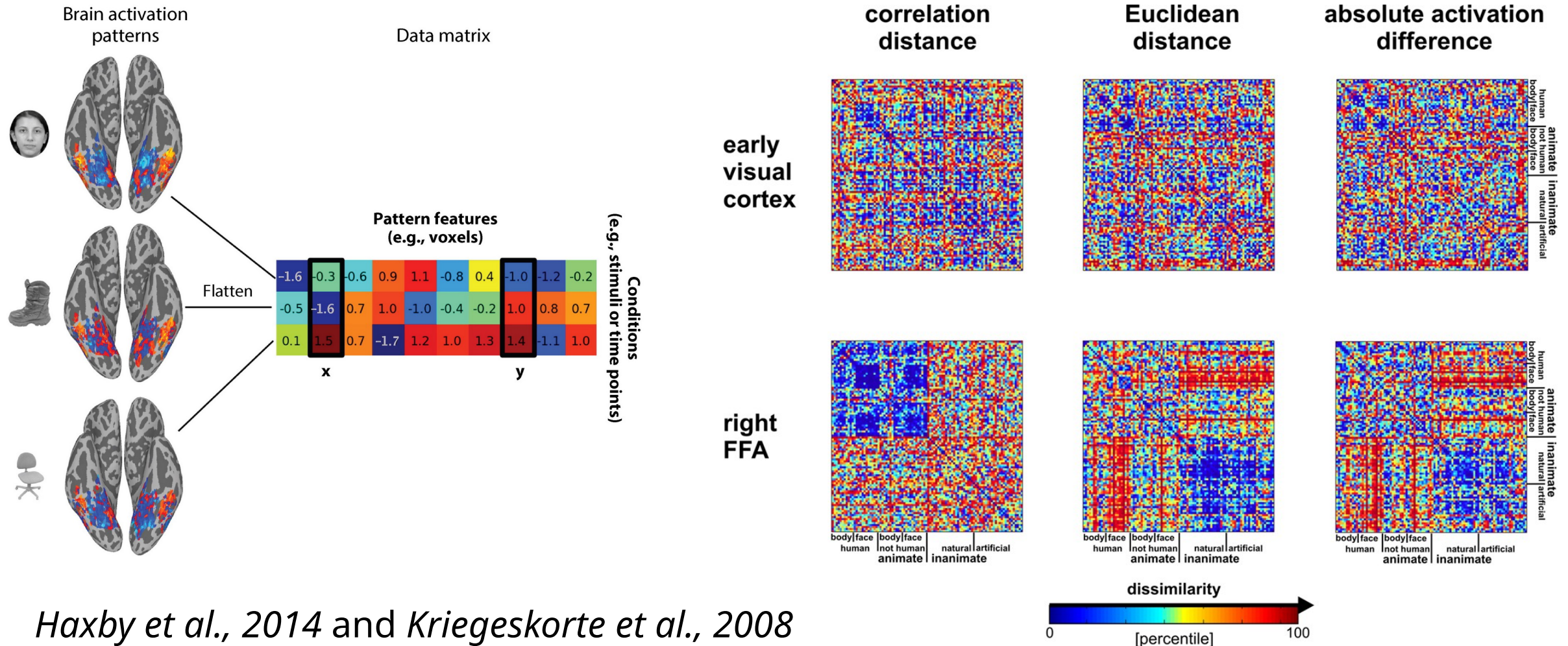
Fig. 2 | RSA-based clustering identifies separate prefrontal and limbic food networks. We applied a network clustering approach based on Representational Similarity Analysis (RSA¹⁵) to examine the representation of food images within food-responsive regions of the brain (A), (see Fig. 1). A We extracted and compared the multivariate food representations of each food from our food-responsive regions-of-interest (ROIs) to generate separate neural Representational Dissimilarity Matrices, B which we then compared across each pair of ROIs, resulting in a second-level ROI similarity matrix reflecting the similarity of representational profiles across our food-responsive regions. C, D We applied a network clustering algorithm to this matrix, which identified an optimal number of 2 clusters and partitioned the ROIs into two networks exhibiting categorically different

representational profiles: a Prefrontal network composed of dorsal brain regions in prefrontal and parietal cortices, and a Limbic network composed of ventral cortico-limbic and sub-cortical brain regions. NB: Node sizes in this graph (D) are proportional to the average similarity of each node to every other node. The coloring of edge lines between nodes indicates whether those edges are within or between networks, and their thickness indicates the edge strength (i.e., similarity). The position of nodes within the graph indicates relative centrality within the whole node network. For copyright reasons, original images have been replaced with visually similar images from the public domain. Photos from Wikimedia.com. See the "Data availability" section for the location of original images.

Avery et al., 2025

RSA clustering (2nd method)

Representational Similarity Analysis (In neurosciences) :



Haxby et al., 2014 and Kriegeskorte et al., 2008

RSA clustering (2nd method)

Representational Similarity Analysis (In our case) :

- Each stimuli is a step of the SSAE task
- fMRI data is our neuron activations

Multiple distances : pearson correlation, covariance, euclidean, cosine, mahalanobis ...

“multivariate noise normalization improves the reliability of all dissimilarity measures”
(*Walther et al. 2015*)

RSA clustering (2nd method)

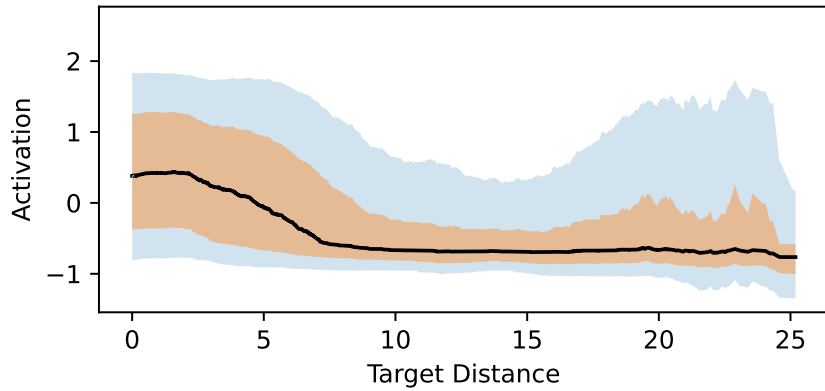
- Clustering our RSA matrix :
 - Hierarchical clustering
 - Graph based community detection (Louvain, Leiden)

Communities detected by the Louvain algorithm

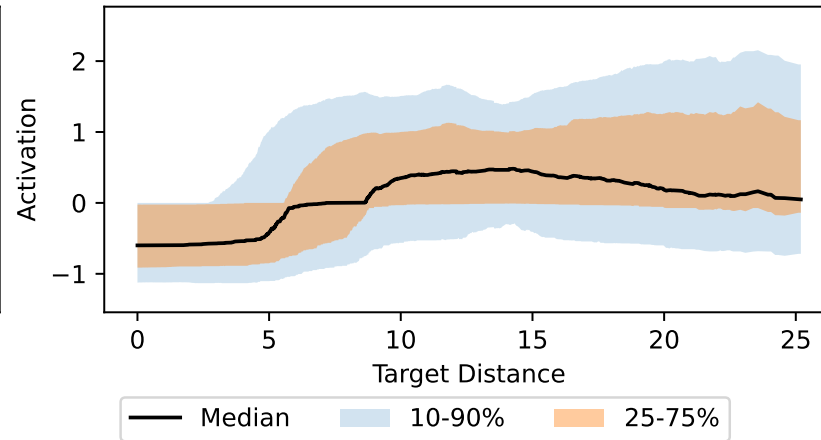


Visualizing neuron clusters

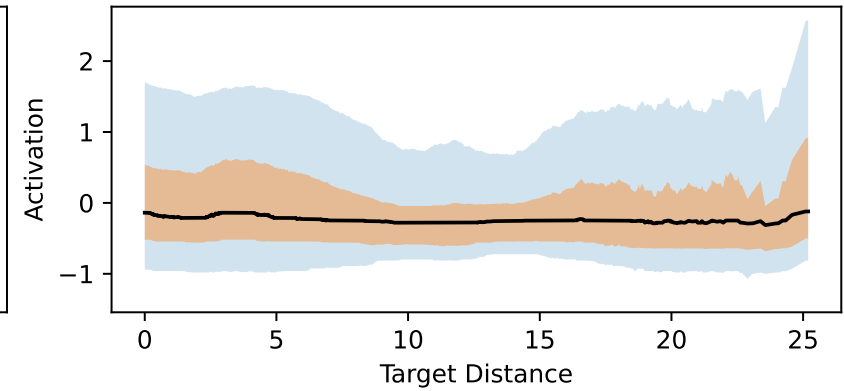
Neuron Cluster: 1



Rolling Neuron Activation vs Target Distance
Neuron Cluster: 2



Neuron Cluster: 3



Decision Transformer

Attention Is All You Need

Why
Their attention
mechanism

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-

Transformer Example (GPTs)

What: Auto-regressively generates text

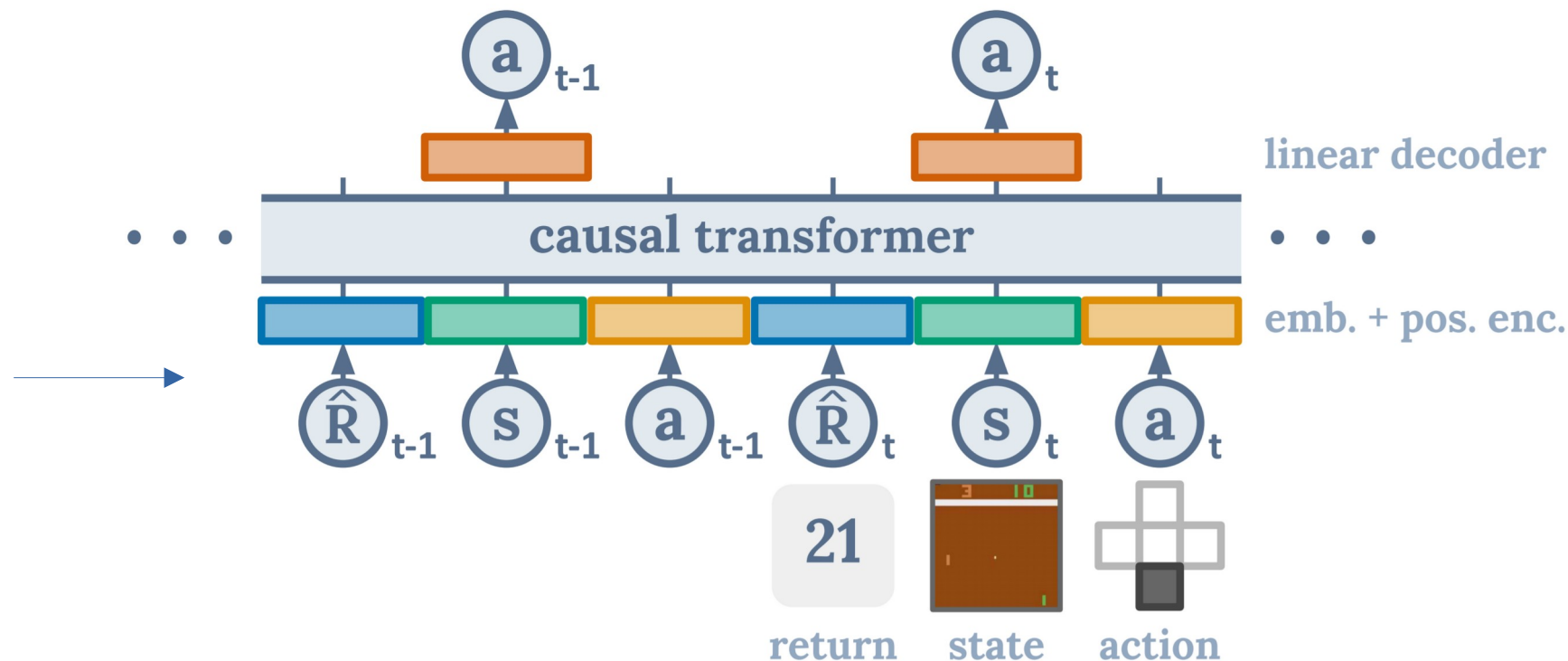
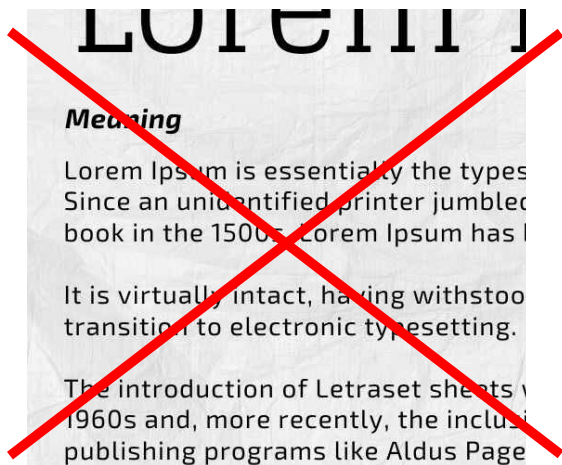
How:

- 1) Takes an input
- 2) Predicts next word
- 3) Appends predicted word to the input
- 4) Repeats

Decision Transformer

How: It doesn't model words but states, returns-to-go and actions

Goal: Predicts next action from previous states, returns-to-go and actions



Chen et al., 2025

Attention mechanism (overview)

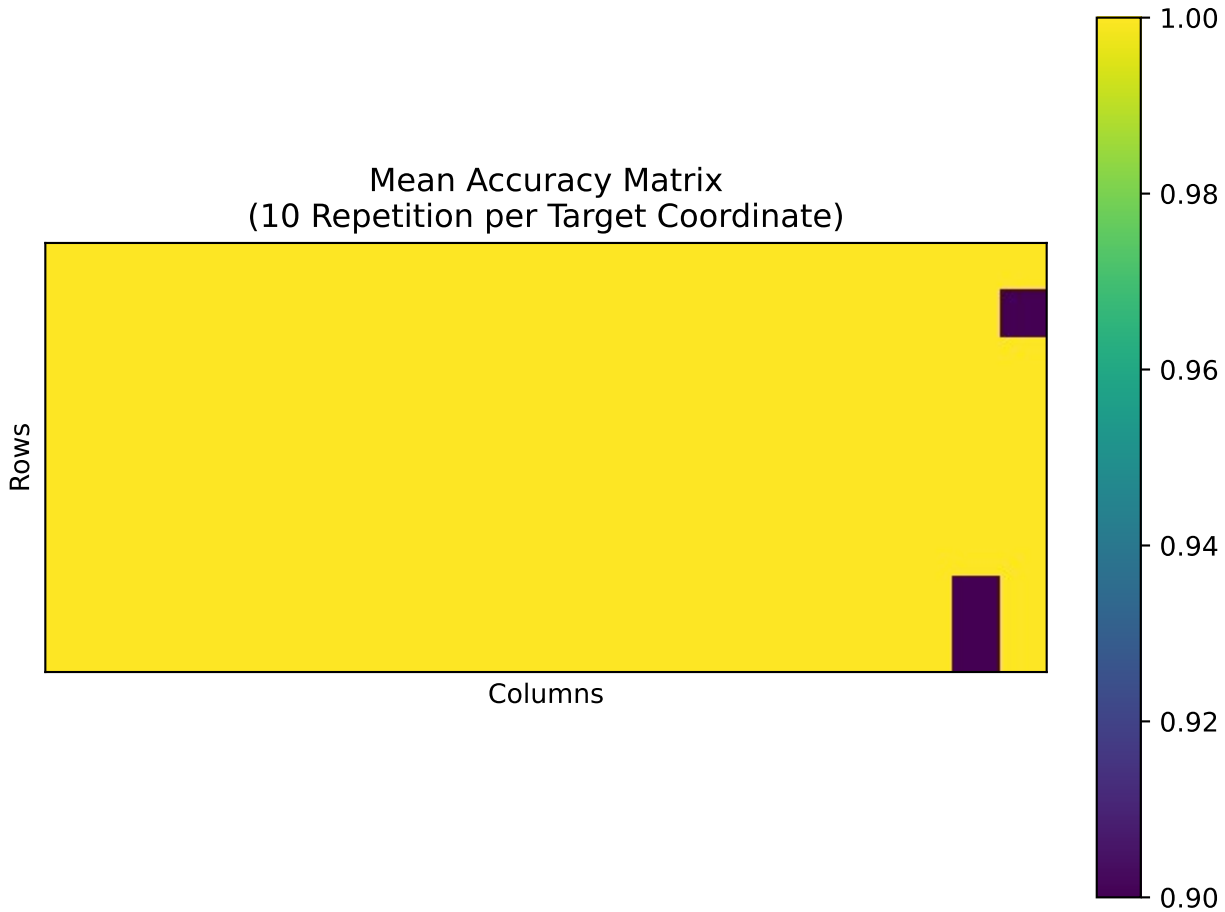
- **With words:**
 - Gives more importance to certain words that have a strong meaning in a certain context, less to others
- **In our case:**
 - Gives more importance to certain states, rtgs or actions
 - For example : the state and action that led to the discovery of the target

Decision Transformer (differences)

- **Differences with the Base Model (PPO):**
 - More complex architecture
 - More computationally intensive
 - Harder to interpret
 - Models attention : weights previous states, rtgs and actions
 - whereas PPO is memoryless
 - Closer to how the brain works (integrates some form of memory)

Decision Transformer (results)

Mean Accuracy Matrix
(10 Repetition per Target Coordinate)



Mean Number of Steps Matrix
(10 Repetitions per Target Coordinate)

